Talank Baral

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PROFILE

Software engineer and Computer Science Ph.D. student working on reliable test automation and efficient continuous integration. More than two years of industry experience maintaining ownCloud acceptance tests and CI pipelines, followed by research on CI build optimization, flaky-test reproduction, regression testing, and AI-assisted software engineering.

RESEARCH INTERESTS

Software Testing · Continuous Integration · Flaky Tests · Regression Test Optimization · LLMs for Software Engineering · Agentic Software Engineering

EDUCATION

2022–Present **Ph.D. in Computer Science** — Fairfax, VA, USA
George Mason University
GPA: 3.98/4.00 • Advisor: Wing Lam

2015–2019 **B.E. in Software Engineering** — Pokhara, Nepal
Gandaki College of Engineering and Science, Pokhara University
GPA: 3.67/4.00

PUBLICATIONS

Conference Papers

[C3] ASE '26 Shanto Rahman, **Talank Baral**, August Shi, and Wing Lam. **TDRepro: A Neurosymbolic Approach to Reproducing Timing-Dependent Flaky Test Failures**. *41st IEEE/ACM International Conference on Automated Software Engineering (ASE 2026)*, pp. 675–686, Munich, Germany, 2026.
ACM SIGSOFT Distinguished Paper Award
Acceptance rate: 8.82% direct acceptance [DOI] [PDF]

[C2] ICSME '26 Atish Kumar Dipongkor, **Talank Baral**, Wing Lam, and Kevin Moran. **Test Coverage Analysis of Agentic Pull Requests**. *42nd International Conference on Software Maintenance and Evolution (ICSME 2026)*, Vision and Emerging Results Track, Benevento, Italy, 2026.
Acceptance rate: 39% [arXiv] [Code] [Artifact]

[C1] ASE '23 **Talank Baral**, Shanto Rahman, Bala Naren Chanumolu, Basak Balci, Tuna Tuncer, August Shi, and Wing Lam. **Optimizing Continuous Development By Detecting and Preventing Unnecessary Content Generation**. *38th IEEE/ACM International Conference on Automated Software Engineering (ASE 2023)*, pp. 901–913, Kirchberg, Luxembourg, 2023.
Acceptance rate: 21% (134/630) [PDF] [Slides]

Tool Demonstrations

[D1] ICSE '25 Demo **Talank Baral**, Emirhan Oğul, Shanto Rahman, August Shi, and Wing Lam. **OptCD: Optimizing Continuous Development**. *47th IEEE/ACM International Conference on Software Engineering (ICSE 2025)*, Demonstrations Track, pp. 45–48, Ottawa, ON, Canada, 2025.
[PDF] [Slides]

Under Review

[U1] **Talank Baral**, Abdelrahman Baz, Atish Kumar Dipongkor, August Shi, Kevin Moran, and Wing Lam. **Using Efficient Test Orders for Regression Testing**. *Under review*.

RESEARCH EXPERIENCE

Aug 2022–Present **Graduate Research/Teaching Assistant** Fairfax, VA, USA
Department of Computer Science, George Mason University

- Developed and evaluated techniques to detect and prevent unnecessary content generation in CI builds (OptCD; ASE 2023 and ICSE 2025 Demonstrations)
- Worked on a neurosymbolic LLM pipeline to reproduce timing-dependent flaky test failures (TDRepro; ASE 2026 Distinguished Paper Award)
- Analyzed test coverage of pull requests authored by AI coding agents (ICSME 2026)
- Developed techniques for efficient test orders in regression testing

INDUSTRY EXPERIENCE

Jan 2020–Aug 2022 **Software Developer** Pokhara, Nepal
JankariTech

- Built and maintained BDD/Gherkin acceptance tests for ownCloud web UI (Nightwatch.js) and desktop synchronization (Squish), alongside API and integration test automation
- Maintained Drone CI pipelines with scheduled nightly runs, chat notifications, and Dockerized test middleware; kept integration suites aligned across ownCloud server, oCIS, and Reva
- Improved failure diagnosis with HTTP and server logs, screenshots on GUI test failure, application crash stack traces, and a GUI test-log parser
- Refactored tests to the Page Object Model; fixed intermittent failures and false-positive tests
- Set up test automation, CI pipelines, and acceptance tests for additional client projects

TECHNICAL SKILLS

CI/CD	GitHub Actions, GitLab CI, Travis CI, Drone, Docker, Maven, Git
Testing	JUnit, Selenium, Cypress, Nightwatch.js, Jest, Puppeteer, Behat, Squish, Cucumber (BDD/Gherkin)
Languages	Python, Java, JavaScript, Bash, SQL, PHP, C, C++

OPEN-SOURCE CONTRIBUTIONS

ownCloud: Contributed web and desktop acceptance tests, CI maintenance, and test-failure diagnostics through JankariTech (professional work described above)

International Dataset of Flaky Tests (IDoFT): Detected and fixed flaky tests in open-source Java projects

CI build optimization: Submitted 25 pull requests to open-source projects that disable generation of unnecessary content in CI builds

pact-cplusplus: Fixed compilation issues in version 0.1.0

avocode-email-tagsinput: Contributed web UI test automation

SELECTED PROJECTS

2023–2025 **OptCD** [Code] [Demo]

Python, Bash, GitHub Actions, Maven, Gemini API

- Instruments a GitHub Actions workflow to find directories generated but never accessed, traces each to the responsible Maven plugin and step, and uses an LLM (Gemini) to propose a fixed workflow that it pushes for validation in CI
- Built the evaluation pipeline: mined Java repositories through the GitHub API, filtered for Maven and GitHub Actions projects, and aggregated file-access logs from 600+ CI job runs into per-job and per-project results
- Findings led to 25 pull requests to open-source projects that disable unnecessary generation

2018–2019 **Gas Station Management System** [Code]

Java, JavaFX, Apache Derby, Maven

- Desktop point-of-sale and accounting app for a fuel station built with JavaFX and an embedded Apache Derby database
- Covers instant and credit billing, credit-company accounts, fuel stock and tank refills, daily closing reports, and receipt printing
- Used at a gas station in Pokhara, Nepal, for more than seven years

HONORS & AWARDS

Oct 2026 **ACM SIGSOFT Distinguished Paper Award**, ASE '26
TDRepro: A Neurosymbolic Approach to Reproducing Timing-Dependent Flaky Test Failures

Aug 2023 **SIGSOFT CAPS Travel Grant**, ACM SIGSOFT
Awarded $1,200 to attend ASE 2023 (declined due to visa constraints)

May 2023 **Summer Research Award**, Department of Computer Science, George Mason University

2015–2019 **Undergraduate Academic Excellence Award**, Gandaki College of Engineering and Science, Pokhara University
Merit-based award

Feb 2018 **First Place, Project Expo**, 7th GCES IT Expo, Pokhara University

Jun 2012 **National Talent Identification Award (First Place Nationwide)**, Government of Nepal
For designing a seed plantation machine

TEACHING

Fall 2026	**SWE 637 / SWE 437**: Software Testing	*Graduate Teaching Assistant*
Spring 2026	**CS 691**: Synergies of LLMs and Software Engineering	*Graduate Teaching Assistant*
Fall 2022	**CS 262**: Introduction to Low-Level Programming	*Graduate Teaching Assistant*

INVITED & CONFERENCE TALKS

May 2025 *OptCD: Optimizing Continuous Development*
ICSE 2025 (Demonstrations Track) • Conference Talk • Ottawa, ON, Canada

Jan 2024 *Optimizing Continuous Development By Detecting and Preventing Unnecessary Content Generation*
ESSEC 2024 • Colloquium Talk • William & Mary, Williamsburg, VA, USA

Jul 2023 *Optimizing Continuous Development*
Microsoft, Redmond Campus • Industry Talk • Redmond, WA, USA

Jul 2023 *Optimizing Continuous Development and Predicting Test Information in Software Development*
ISSTA 2023 • Doctoral Symposium • Seattle, WA, USA